Parand Khalili

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EDUCATION

Amirkabir University of Technology *Sep 2022 - Present*
BSc in Computer Engineering GPA: 18.47/20
Thesis: A Structure-Aware LLM-Based Framework for Graph Node Labeling

RESEARCH INTERESTS

- **Data Analytics & Data Mining:** Real-world data analysis, pattern discovery, and data-driven insights.
- **Graph Machine Learning:** Graph representation learning, GNNs, and node classification.
- **Large Language Models:** LLM–GNN integration and structure-aware learning on text-attributed graphs.
- **Data-Efficient Learning:** Active learning, automated graph labeling, and confidence-aware pseudo-labeling.
- **Explainable AI (XAI):** Model interpretability, explainable graph learning, and transparent AI decision-making.

PROFESSIONAL EXPERIENCE

Retail Data Analyst | EMRC *Jul 2026 – Present*

- Analyze large-scale, real-world retail datasets and deliver client-facing reports with actionable business insights.
- Developed a Python application to automate a repetitive internal workflow, replacing manual steps with a reusable process.
- Currently developing a customizable, end-to-end automation pipeline for generating reports tailored to clients' requirements.

Data Analysis & BI Intern | YAS Arghavani System Engineering Co. (Bank Mellat IT Holding) *Summer 2025*

- Developed automated Power BI dashboards to monitor POS performance, maintenance, and operational efficiency.
- Processed and modeled transactional data using Power Query and DAX to generate KPI-driven revenue insights.
- Integrated CRM and payment systems to improve reporting clarity and cross-functional access to business insights.

TEACHING ASSISTANT EXPERIENCES

Head Teaching Assistant – Business Intelligence *Fall 2025*
Instructor: Dr. Ahmadi

- Designed instructional materials and course projects.
- Taught Power BI, Python, SQL, and data modeling for data analysis.

Teaching Assistant Roles

- **Data Mining** | Dr. Nazerfard *Spring 2026*
- **Advanced Software Engineering** | Dr. Gohari *Spring 2026*
- **Theory of Languages & Automata** | Dr. Zakeri *Fall 2025*
- **Software Engineering** | Dr. Alirezaei *Fall 2025, Spring 2025*
- **Applied Linear Algebra** | Dr. Haghirchehraghani, Dr. Nazerfard *Fall '24, Spring '25, Fall '25*
- **Microprocessor & Assembly Language** | Dr. Farbeh *Fall 2025*
- **Computer Architecture** | Dr. Farghadan *Spring 2025*
- **Advanced Programming** | Dr. Zeinali *Spring 2024*

CERTIFICATIONS

Data Analysis Certificate | Quera *Summer 2025*

- Completed practical projects in data cleaning, transformation, and exploratory analysis using Pandas and NumPy.

Advanced Python Bootcamp | Sematec *Sep 2024 – Dec 2024*

- Built a Python–SQL desktop app applying OOP, design patterns, and core software engineering principles.

HONORS AND AWARDS

National Countrywide University Entrance Exam (Konkour), Iran *2022*
Ranked 134 among over 150,000 participants nationwide.

NOTABLE PROJECTS

BSc Thesis: A Structure-Aware LLM-Based Framework for Graph Node Labeling *2026 – Present*

- Conducted research in Dr. Mostafa Haghir Chehreghani's lab, which developed into my undergraduate thesis.
- Developing a label-efficient node classification framework that combines LLM-generated annotations with Graph Neural Networks on text-attributed graphs.
- Investigating structure-aware prompts using 1-hop/2-hop neighbors and random walks, active node selection, and confidence-based filtering.
  Technologies: Python, PyTorch, PyTorch Geometric (PyG), LLMs, GNNs.

Credit Scoring & Loan Default Risk Mitigation Framework *2026*

- Built an end-to-end ML pipeline with target encoding and engineered financial metrics to handle class imbalance.
- Tuned a Random Forest model achieving 87.41% accuracy, 79.21% balanced accuracy, and a 4.65 decile lift.
  Technologies: Python, Scikit-Learn, Advanced ML Metrics.

Tetris AI via Genetic Algorithms *2025*

- Created an autonomous Tetris agent using evolutionary strategies (Selection, Crossover, Mutation).
- Optimized a fitness function for board evaluation and visualized the training process.
  Technologies: Python, Pygame, Genetic Algorithms.

Fundamentals of Artificial Intelligence Portfolio *2025*

- Built AI agents for Sokoban (A*), Isolation (Minimax/Alpha-Beta), and Crossword (CSP/AC-3).
- Implemented Reinforcement Learning (Q-Learning) and Bayesian Networks (Gibbs/Likelihood Weighting).
  Technologies: Python, NumPy, Matplotlib, Genetic Algorithms.

MyTorch Framework *2025*

- Implemented core components: tensors, backpropagation, activation functions, and SGD optimizer.
  Technologies: Python, PyTorch, Linear Algebra, Neural Networks.

xv6 Operating System Enhancements *2025*

- Added custom system calls and implemented multithreading capabilities to the xv6 kernel.
- Managed memory and process scheduling in a C-based Linux/Docker environment.
  Technologies: C, Linux, Shell Scripting, Docker.

SKILLS AND QUALITIES

Programming Languages: Python, C++, C, Java

Machine Learning & Graph AI: Graph Neural Networks (GCN, GAT), LLM-GNN Integration, Active Node Classification, Ensemble Methods, Structure-Aware Graph Labeling, Active Learning

Tools and Ecosystem: PyTorch, PyTorch Geometric (PyG), NumPy, Pandas, Scikit-learn, Matplotlib, SQL database

Business Intelligence: Power BI (DAX, Power Query, Power Pivot), HR & organizational analytics

DevOps and Tools: Git, Linux, Docker

Languages: English (Full Professional Proficiency), Persian (Native)
TOEFL iBT (1–6 scale): Reading: 5/6 | Listening: 6/6 | Writing: 5/6 | Speaking: 4.5/6